DAY-2

- **Introduction to Dynamic Routing Theory : What is dynamic routing? Comparison between static and dynamic routing. Practical -Observe RIP routing updates in a pre-configured network.**

1. List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

ANS:

Feature	Static Routing	Dynamic Routing	Real-World Analogy
Route Configuration	Routes are entered manually by the network administrator.	Routes are learned and updated automatically using routing protocols like RIP.	**Google Maps:** Static routing is like following a fixed route you planned before leaving. Dynamic routing is like Google Maps changing your route automatically when there is traffic.
Network Changes	Does not adapt automatically if a network link fails.	Automatically finds an alternative path when a link fails.	**Swiggy Delivery:** A delivery partner following one fixed road even if it is closed is static routing. If the app changes the route because of roadblocks, it is dynamic routing.
Maintenance	Requires manual updates whenever the network changes.	Requires little manual work because routers exchange routing information automatically.	**Uber:** Manually telling the driver every turn is like static routing. Uber automatically selecting the fastest route is like dynamic routing.

2. Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

ANS:

- **Router R1**
  - **show ip route**

Destination Network	Learned Via	Metric (Hop Count)
192.168.2.0/24	RIP	1
192.168.3.0/24	RIP	2

  - **show ip protocols**
    - Routing Protocol: **RIP**
    - Version: **RIP Version 2**
    - Update Timer: **30 seconds**
    - Invalid Timer: **180 seconds**
    - Hold-down Timer: **180 seconds**
    - Flush Timer: **240 seconds**

- **Router R2**
  - **show ip route**

Destination Network	Learned Via	Metric
192.168.1.0/24	RIP	1
192.168.3.0/24	RIP	1

  - **show ip protocols**
    - Routing Protocol: **RIP**
    - Version: **RIPv2**
    - Update Timer: **30 seconds**
    - Invalid Timer: **180 seconds**
    - Hold-down Timer: **180 seconds**
    - Flush Timer: **240 seconds**

- **Router R3**
  - **show ip route**

Destination Network	Learned Via	Metric
192.168.1.0/24	RIP	2
192.168.2.0/24	RIP	1

  - **show ip protocols**
    - Routing Protocol: **RIPv2**
    - Update Timer: **30 seconds**
    - Invalid Timer: **180 seconds**
    - Hold-down Timer: **180 seconds**
    - Flush Timer: **240 seconds**

3. Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route. Hint : Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

ANS:

- **Step 1: Shut down an interface**
  Router(config)# interface serial0/0/0
  Router(config-if)# shutdown

- **Step 2: Check routing table repeatedly**
  show ip route

- **Observation Table**

Event	Observation
Interface Shutdown	The connected network became unreachable.
RIP Update Sent	Neighbor routers received the updated routing information.
Routing Table Updated	The unreachable route was removed automatically.
Approximate Time Taken	**About 180 seconds (3 minutes)** for the route to become invalid and be removed from the routing table.

4. Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.

ANS:

- **Uber/Zomato**

Dynamic routing in computer networks works similarly to apps like Uber or Zomato. When a road is blocked or traffic increases, these apps automatically calculate a faster route for the driver without requiring manual changes. RIP also updates routing information automatically when a network path fails and selects another available route if one exists. In both cases, the goal is to find the best available path and ensure that the destination is reached efficiently.

- **Introduction to Dynamic Routing**

Dynamic routing is a method in which routers automatically exchange routing information using routing protocols such as RIP, OSPF, or EIGRP. These protocols allow routers to learn the best available paths to different networks and update their routing tables whenever the network changes. Dynamic routing reduces manual configuration and is suitable for medium and large networks where routes frequently change.

- **What is Dynamic Routing?**

Dynamic routing is the process of automatically discovering, maintaining, and updating network routes using routing protocols. Routers communicate with each other to share routing information and choose the best path for forwarding data packets.

- **Comparison Between Static Routing and Dynamic Routing**

Static Routing	**Dynamic Routing**
Configured manually by the administrator.	Learned automatically using routing protocols.
Does not adapt automatically to network failures.	Automatically updates routes when the network changes.
Suitable for small networks.	Suitable for medium and large networks.
Requires more manual maintenance.	Requires less manual maintenance after configuration.
No routing protocol is used.	Uses protocols such as RIP, OSPF, EIGRP, or IS-IS.